Bhanu Prasad Golla

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SUMMARY

MSc Embedded Systems Design student with nearly three years of experience in automotive software and system testing, requirements-based validation, and debugging for the vehicle powertrain and autonomous functions. Experienced in analysing system requirements, developing structured test cases and test plans, executing manual and automated tests, and investigating system-level issues. Hands-on experience with Python, C, MATLAB/Simulink, Stateflow, IBM DOORS, JIRA, Confluence, and CI/CD workflows. Strong interest in automotive ECU system development, E/E architectures, vehicle communication, diagnostics, and software-defined vehicle platforms.

SKILLS

Programming: (Python, C, Basic C++) | **Development & Project Tools:** (JIRA, Confluence, IBM DOORS, Git/CI/CD) | **Software Development Methods:** (Agile/Scrum and Cross functional collaboration) | **Automotive Software & Architecture:** (AUTOSAR- Basic knowledge, E/E architecture fundamentals, ECU system development) | **Automotive Communication Protocols:** (CAN, LIN, FlexRay and Ethernet) | **Functional Safety and Quality:** (ISO 26262, ASPICE, FMEA and Fault Tree Analysis)

PROFESSIONAL EXPERIENCE

TATA ELXSI, *Test Engineer - Automotive Systems* 07/2022 – 04/2025 | Hyderabad,India

- Analysed vehicle-feature, software and system-level requirements for Powertrain and ADAS functions and translated them into structured test cases, test sets and validation plans.
- Developed requirements-based test cases and test strategies to verify automotive system behaviour against defined functional requirements.
- Executed manual and automated system tests and evaluated actual system behaviour against expected results.
- Developed and maintained keyword-driven test scripts and supported test automation activities for vehicle-level testing environments.
- Investigated failed test results, analysed deviations and supported debugging and root-cause investigation of software and system-level issues.
- Performed CI/CD pipeline checks and quality reviews to improve test completeness, consistency and software-quality assurance.
- Created and maintained test reports, validation results, technical documentation and traceability information throughout the development lifecycle.
- Worked within an Agile/Scrum multidisciplinary development environment, using JIRA and Confluence for technical documentation, issue management, task tracking and project-status reporting.

Nucleonix Systems Pvt.Ltd, *Embedded Intern* 10/2021 – 02/2022 | India

- Supported the development and testing of embedded measurement systems and microcontroller-based devices.
- Performed functional testing, sanity checks, calibration, and validation of completed systems.
- Debugged circuit and system-level issues and supported troubleshooting of hardware/software integration problems.

- Assisted with microcontroller programming and PCB-level testing during product development.
- Performed quality checks to verify correct system operation before final delivery.
- Prepared technical documentation and user manuals for developed systems.

PROJECTS

Autonomous Navigation of Lego Ev3
- Analysed software and system requirements and translated them into functional control behaviour for an embedded robotic platform.
- Developed model-based control software using MATLAB/Simulink and Stateflow for straight, circular and square movement scenarios.
- Created structured test cases, test plans and test harnesses for functional verification of system requirements.
- Performed simulation, Software-in-the-Loop testing, debugging and validation before deployment to physical hardware.
- Generated embedded code, deployed the validated model to the EV3 target and performed final system-level hardware testing.
- Prepared complete technical, testing and validation documentation.

FPGA-Based Stepper Motor Control and Monitoring System
- Developed embedded software in C for motor configuration, interrupt-based sampling, UART communication, and data handling.
- Developed a Java/Eclipse GUI for speed, direction, and position control.
- Integrated software and hardware components and performed functional and integration testing.
- Debugged software/hardware communication and analysed system behaviour.
- Validated the complete system through simulation and hardware testing.
- Documented software architecture, interfaces, and verification results.

Acceleration-Based Elevator Height Measurement System
- Designed, assembled, and tested an electronic acceleration measurement circuit using an ADXL335 MEMS accelerometer.
- Developed analog signal-conditioning circuitry including offset compensation, amplification, and active low-pass filtering.
- Configured the ATmega32 microcontroller for ADC-based sensor data acquisition and serial communication.
- Performed functional testing, calibration, circuit debugging, and measurement analysis using laboratory equipment.
- Analysed sensor noise, calibration errors, and signal deviations and documented the complete circuit design and test results.

Safety and Reliability Analysis of a Car Braking System
- Performed a Functional Hazard Assessment for a dual-circuit hydraulic automotive braking system and classified potential hazards according to severity.
- Conducted Failure Mode and Effects Analysis (FMEA) covering mechanical, hydraulic and electrical system components.
- Developed fault-tree models to analyse system-level failure combinations and calculate failure probabilities.
- Evaluated safety-critical single points of failure and defined quantitative reliability requirements for total braking-system failure.
- Identified the tandem master cylinder as a critical system element and evaluated additional braking mechanisms as potential risk-mitigation measures.

- Documented safety assumptions, failure behaviour, reliability calculations and proposed mitigation strategies.

Discrete Control of a Seesaw System 05/2025 – 08/2025

- Developed a nonlinear mathematical model of a seesaw–cart system in MATLAB/Simulink using its mechanical parameters and equations of motion.
- Linearized the nonlinear equations around the zero operating point and derived a four-state state-space model.
- Compared linear and nonlinear model responses and identified that the linear approximation is accurate only near the selected operating point.
- Observed significant divergence between the models after approximately 1–1.5 seconds for larger state deviations.
- Designed a full-state feedback controller using Ackermann's pole-placement method, including state-feedback and preamplifier gains.
- Investigated different pole locations and designed an augmented fifth-order PI state-feedback controller to improve disturbance rejection and steady-state accuracy.

Moving Light Controller on ZedBoard 04/2025 – 07/2026

• Implemented an 8-bit sequential LED pattern controller with pushbutton-controlled left/right rotation, stop and pattern-load functions.
• Developed a clock-divider process and synchronous control logic; validated via testbench simulation in Vivado and on ZedBoard hardware.

Hardware-Accelerated 8-Bit PRBS Generator on Zynq-7000 04/2025 – 06/2025

• Designed an 8-bit PRBS generator core in VHDL using a Fibonacci LFSR and packaged it as a custom AXI4-Lite slave IP core.
• Integrated the IP with the ARM Cortex-A9 processing system via AXI interconnect; mapped ZedBoard switches/pushbuttons and LEDs as I/O.
• Developed a bare-metal C application in Vitis with an interactive serial command interface for seed loading and step control, verified via KiTTY.

EDUCATION

Hochschule Bremerhaven 04/2025 – Present | Bremerhaven, Germany
Masters in Embedded System Design

Veltech University, *Bachelors in Electrical and Electronics Engineering* 06/2019 – 04/2022 | Chennai,India

Jayapraksh Narayan Engineering College 06/2016 – 03/2019 | Mahabunagar, India
Diploma in Electrical and Electronics Engineering

LANGUAGES

English - C1 | German - B1